

BS-CAT Model — Liability Module Performance Report

OOT Sample | 1,519 Applications | 1,958,480 Transactions | Submission Date: 2026/07/27–2026/07/31

1. Summary

The BS-CAT classification taxonomy remains fully aligned with Illion's framework. Every category label that Illion produces has a direct counterpart in BS-CAT's output, **ensuring that a migration from Illion to BS-CAT will have minimal impact on downstream systems that consume these labels.**

We continue to focus on the **Liability Module**, which has completed **two rounds of question-and-answer (Q&A) review** against manual human review, and **the model has successfully addressed the issues raised during that review.** The remaining discrepancies stem from objective data limitations — specifically, truncated bank-statement windows — rather than model errors.

The current benchmark now includes an additional fresh week of out-of-time (OOT) data, spanning 1,519 cases and a total of 1,958,480 bank transaction lines. **It is worth noting that this comparison has been run across a large number of test samples.** Across this extended evaluation, BS-CAT's loan detection coverage reached 12.83%, modestly exceeding Illion's 12.31%. These results show that BS-CAT does not underperform Illion in loan detection coverage. Among jointly detected loans, classification agreement reaches 90.7%, **confirming strong overall alignment between the two systems.**

A dedicated Dishonours assessment has been added to this edition. BS-CAT achieves 100% recognition of all Illion-flagged Dishonours. Among transactions that both systems classify as Dishonours and assign a specific counterparty, counterparty agreement is 95.2%. The 4.8% discrepancy is confined to non-lender entities (e.g. YOUI, Suncorp, Bupa, Telstra), where BS-CAT correctly refrains from forcing a lender counterparty, as the debit initiator is not a lending institution.

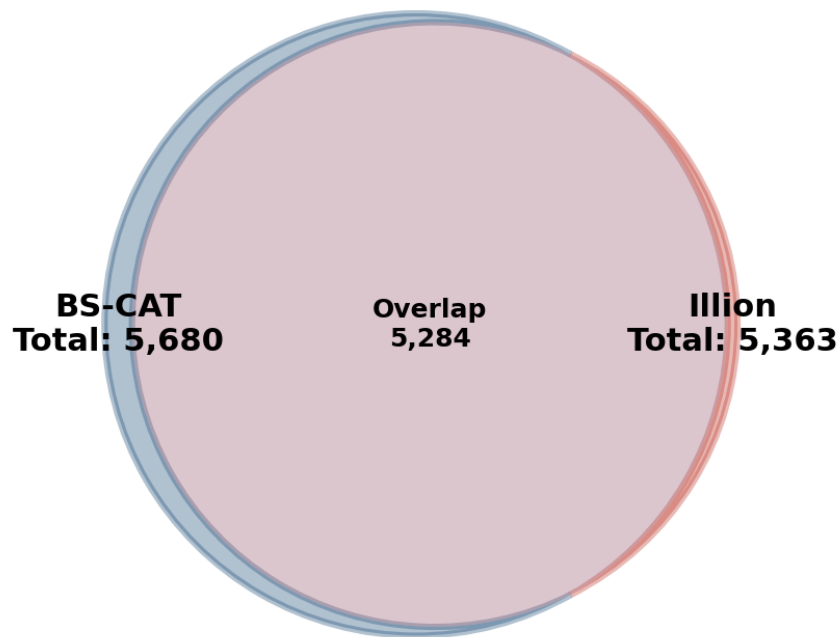
2. Loan Detection Coverage

2.1 Coverage and Cross Matrix

Out of 1,958,480 bank transaction records, BS-CAT identifies 251,273 (12.83%) as loans, while Illion identifies 241,089 (12.31%). BS-CAT's coverage rate exceeds Illion's by 0.5 percentage points.

The coverage overlap between the two systems is illustrated below:

Loan Detection Coverage Overlap



BS-CAT exclusive detections: 396 records (0.9%); Illion exclusive detections: 79 records (0.2%). Both systems have detection gaps, but BS-CAT misses fewer loans and delivers superior overall coverage.

2.2 BS-CAT Detected, Illion Missed (396 records)

BS-CAT exclusively detected 396 loan transactions. Of these, 327 received no category label from Illion, predominantly in BNPL and EWA products:

(A) Illion provided no category label (327 records):

Sample Text	Count	BS-CAT Category	BS-CAT Counterparty	Illion Category	Illion Third Party
TRANSFER CREDIT PARK TERRACE MOTEL PTY LT COON 2026-06-20 3 of 4	326	Non SACC Loans	NAB Now Pay Later	-	-
TRANSFER CREDIT LESLEY KENNEDYLesley Afterpay	1	Non SACC Loans	Afterpay	-	-

(B) The remaining 69 records were labeled by Illion as non-loan categories (e.g., Fees, Internal Transfer) but correctly identified as loans by BS-CAT. In these cases, BS-CAT's loan product transaction pattern recognition successfully captured loans that Illion missed.

Sample Text	Count	BS-CAT Category	BS-CAT Counterparty	Illion Category	Illion Third Party
Direct Credit 421520 DOT Express Pty DOT Express Pay	21	Non SACC Loans	PressPay	Wages	DOT Express Pty DOT Express Exps
Humm90 Mastercard G	12	Non SACC Loans	Humm	Credit Card Repayments	Humm90 Credit Card Repayment
Loan A/C Fee	11	Non SACC Loans	Generic Loans	Fees	-
Return 14/05/26 Loan Repayment LN REPAY	6	Non SACC Loans	CBA Loans	Dishonours	CBA Loans

474279699 Value Date: 14/05/2026					
MISCELLANEOUS DEBIT V7631 20/06 WOOLWORTHS/23 STEEL STREENEWCASLE 74278246171	4	Unknown Loans	Newcastle	Groceries	Woolworths
NIB QANTAS - Visa Purchase - Receipt 183692In NEWCASTLE Date 16 Mar 2026 Card 46	3	SACC Loans	Newcastle	Insurance	Qantas Assure
EFTPOS DEBIT 0406667 Souvlaki GR Sunshine Sunshine West 06/02	3	SACC Loans	Sunshine Loans	Dining Out	Miscellaneous Greek Cuisine
Home loan - Receipt 174703 - To Orange Everyday	2	Non SACC Loans	Home Loan	Internal Transfer	Internal Transfer Debit
DEPOSIT-OSKO PAYMENT 2892856 ETIKA CEINATURAGA soli	2	SACC Loans	Etika Australia	All Other Credits	Other Credits
TRANSFER CREDIT NimbleCT.1ri2c0CT.1ri2c0 NIMBL 2590	1	Non SACC Loans	Nimble	External Transfers	Miscellaneous Funds Transfer
Fundo Loans Surry Hills AU	1	SACC Loans	Fundo Loans	Information	-
EFTPOS DEBIT 0271669 CITI MART PTY LTDCIT I MELBOURNE 09/05	1	Unknown Loans	Citibank Loans	Groceries	Miscellaneous Grocery stores
Too Good To Go including -3.01 Round-up	1	Unknown Loans	Good To Go	Groceries	Groceries
DEBIT CARD PURCHASE HJs Sunshine Sunshine AUS	1	Unknown Loans	Sunshine Loans	Dining Out	Dining Out

Illion shows clear detection blind spots for BNPL, EWA, and Home Loan products. BS-CAT's coverage is materially higher in these emerging credit categories.

2.3 Illion Detected, BS-CAT Missed (79 records)

Illion identified 79 transactions as loans (SACC Loans / Non-SACC Loans) that BS-CAT did not classify as loans. These fall into the following categories:

(1) Credit Card Repayments (29 records)

Illion Category	Illion Third Party	BS-CAT Category	BS-CAT Counterparty	Sample Text	Count
Non SACC Loans	WESTPAC Credit Card Cash Advance Transfer	Credit Card Repayments	WESTPAC Credit Card Cah Advance	CASH ADVANCE Z29868 CASH ADVANCE ON LINE	29

Root cause — Classification rule differences: BS-CAT and Illion apply different classification rules for WESTPAC credit card cash advances. BS-CAT categorizes these as Credit Card Repayments (since a cash advance is essentially a repayment toward the credit card balance), while Illion classifies them as Non-SACC Loans. Both approaches have valid rationale; this is a rule-set difference rather than a model performance gap.

(2) Retail and Debt Collection (7 records)

Illion Category	Illion Third Party	BS-CAT Category	BS-CAT Counterparty	Sample Text	Count
Non SACC Loans	Cash Converters	Retail	Cash Converters Retail	MISCELLANEOUS DEBIT V8178 13/02 CASH CONVERTERS Perth 741860	6
Non SACC Loans	Credit Corp Loan	Debt Collection	Debt Collection	RTGS HIGH VALUE PAYMENT REF NO	1

				1147561 CREDIT CORP GROUP WBC	
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Root cause — Classification rule differences: Cash Converters and Credit Corp each operate multiple lines of business: Cash Converters runs both retail/pawn and micro-lending services, while Credit Corp operates consumer lending and debt collection. Illion applies a blanket Non-SACC Loans label to all transactions from these entities (institution-level categorization), whereas BS-CAT leverages liability product data to distinguish among business lines within the same institution: Cash Converters retail transactions are classified as Retail (6 records), and Credit Corp debt collection is classified as Debt Collection (1 record). This demonstrates BS-CAT's finer-grained recognition capability in multi-business-line scenarios.

(3) Unclassified (Empty) (43 records)

Illion Category	Illion Third Party	BS-CAT Category	BS-CAT Counterparty	Sample Text	Count
Non SACC Loans	Zip Pay	-	-	Zip	2
SACC Loans	Online Credit 365	-	Zainah Pty Ltd	AUTOMATIC DRAWING Online Credit 365 Zai Personal Account	4
Non SACC Loans	Credit Corp Loan	-	-	WITHDRAWAL MOBILE 4060310 BPAY CCFS	10

Root cause — Knowledge base gaps: The counterparty field is empty for these transactions, so the liability engine cannot link them to liability product records. The merchant knowledge base also lacks finv_category mappings for entities such as Zip Pay, MoneyMe, MoneyPlus, CBA Loans, JLM HOME LOANS, and Online Credit 365, resulting in an empty final classification. The merchant_kb and liability product-to-counterparty mappings need to be expanded to cover these entities.

2.4 Summary

BS-CAT outperforms Illion in loan detection coverage (5,680 vs. 5,363), with exclusive detections concentrated in BNPL and EWA products — areas where Illion has clear coverage gaps. Although both systems miss some loans in cross-validation, BS-CAT's under-detection rate is lower. The immediate optimization priority is to bulk-expand the institution and product knowledge base, which would enable near-complete capture of the currently missed loans and further strengthen overall detection coverage.

3. Classification Agreement

3.1 Agreement Rate and Cross Matrix

The following analysis focuses on the 5,284 (11.9%) transactions jointly identified as loans by both systems, comparing label-level agreement. The Unknown Loans category arises primarily from time truncation in bank statement data: when the disbursement transaction falls outside the observation window, the model can only observe repayments without the corresponding drawdown, and conservatively defaults to Unknown.

The label migration cross matrix by Illion category is presented below:

	BS-CAT SACC	BS-CAT Non-SACC	BS-CAT Unknown	Illion Subtotal
Illion SACC	262 (60.0%)	91 (20.8%)	80 (18.3%)	437
Illion Non-SACC	106 (2.2%)	4601 (93.4%)	144 (2.9%)	4926
BS-CAT Subtotal	377	5072	224	5284

Diagonal entries (262 + 4,601) = 4,863 records with full label agreement.

Among the 5,284 jointly detected loans, the classification agreement rate is 92.0%, the disagreement rate is 3.7%, and the Unknown rate is 4.2%.

Illion exhibits a notable over-classification tendency on the SACC category. Among Illion-labeled SACC Loans, nearly 40% are reclassified by BS-CAT — 20.8% migrate to Non-SACC Loans and 18.3% to Unknown Loans, totaling 39.1% divergence. In contrast, Illion's Non-SACC Loans show 93.4% agreement with BS-CAT, with only 2.2% reclassified to SACC and 2.9% to Unknown. This stark asymmetry indicates that Illion over-applies the SACC label, while BS-CAT corrects this bias through loan-stream matching and product information verification, falling back to Unknown only when evidence is genuinely insufficient. BS-CAT's classification logic is demonstrably more precise and robust.

3.2 Disagreement Analysis

3.2.1 Cause 1: Multiple Concurrent Loan Streams at the Same Institution

A single borrower may hold multiple loan facilities of different sizes at the same institution. BS-CAT operates at the loan-stream level by matching disbursements to repayments, enabling it to distinguish parallel loan streams and correct Illion's intra-stream labeling inconsistencies.

Case 1: Application ID=2536067, MoneySpot, loan stream sacc_001

This borrower has 72 transactions at MoneySpot across multiple loan streams. Focusing on loan stream sacc_001, out of 11 transactions, Illion labels the first 6 as SACC Loans and the remaining 5 as Non-SACC Loans — contradicting itself within the same stream. BS-CAT consistently labels all 11 as SACC Loans:

Application ID	Date	Amount	Dr/Cr	Institution	Illion Category	BS-CAT Category	Mismatch
2536067	2026-04-14	\$750.00	credit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-04-22	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-04-29	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-06	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-13	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-20	\$49.09	debit	MoneySpot	SACC Loans	SACC Loans	
2536067	2026-05-27	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-03	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-10	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-17	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes
2536067	2026-06-24	\$49.09	debit	MoneySpot	Non SACC Loans	SACC Loans	Yes

3.2.2 Cause 2: Product Classification Boundary Differences

Certain lending institutions offer LOC or auto loan products that differ materially from SACC Loans in regulatory treatment, interest rate caps, and risk management. Where transaction patterns more closely resemble LOC or term auto loans, Illion defaults to a blanket SACC Loans label. BS-CAT provides finer-grained sub-categorization (e.g., LOC, Car Loan) within Non-SACC Loans.

Case 2: Application ID=2536017, Credit24

Across 19 Credit24 transactions under this application, Illion labels all as SACC Loans. BS-CAT, after loan-stream decomposition, determines the pattern is more consistent with a line of credit (LOC) product, and classifies them as Non-SACC Loans with the LOC sub-category — better reflecting the underlying business nature.

Application ID	Date	Amount	Dr/Cr	Institution	Illion Category	BS-CAT Category	Mismatch
2536017	2026-01-27	\$140.71	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-01-30	\$105.02	debit	Credit24	SACC Loans	Non SACC Loans	Yes

2536017	2026-02-05	\$42.75	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-02-13	\$105.07	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-02-27	\$105.07	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-03-09	\$82.64	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-03-13	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-03-27	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-04-10	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-04-24	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-04-30	\$168.54	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-05-08	\$105.04	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-05-22	\$105.04	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-01	\$83.57	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-05	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-19	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-06-29	\$83.26	credit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-07-03	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes
2536017	2026-07-17	\$105.05	debit	Credit24	SACC Loans	Non SACC Loans	Yes

3.2.3 Cause 3: Conservative Unknown Loan Strategy

An additional source of divergence is BS-CAT's Unknown Loans classification, accounting for 224 records (4.2%) among jointly detected loans.

Case 3: Application ID=2534635, Fundo Loans

This borrower has 12 transactions at Fundo Loans (including 2 dishonors), all belonging to a single loan stream. With no disbursement record present in the bank statement data, BS-CAT cannot definitively determine SACC vs. Non-SACC and labels all as Unknown Loans.

Application ID	Date	Amount	Dr/Cr	Institution	Illion Category	BS-CAT Category
2534635	2026-04-23	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-07	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-21	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-22	\$70.00	credit	Fundo Loans	Dishonours	Dishonours
2534635	2026-05-22	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-05-23	\$70.00	credit	Fundo Loans	Dishonours	Dishonours
2534635	2026-05-30	\$105.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-06-02	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-06-04	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-06-12	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-07-03	\$70.00	debit	Fundo Loans	SACC Loans	Unknown Loans
2534635	2026-07-11	\$50.00	debit	Fundo Loans	SACC Loans	Unknown Loans

4. Dishonours Analysis

4.1 Dishonours Coverage

Our BS-CAT model achieved **100%** identification of all Dishonours flagged by Illion, with zero omissions. **The total coverage volume is slightly higher than that of Illion.** The table below details the Dishonours uniquely identified by BS-CAT.

Text Sample	Count	Illion Category	BS-CAT Category
REVERSAL CREDIT REVERSAL OF DEBIT CREDIT REPAIR AUSTRA004149...	2	Debt Consolidation	Dishonours

Both transactions contain “REVERSAL CREDIT”, which represents a reversal entry resulting from a failed debit. Our model identified these as Dishonours, while Illion classified them under Debt Consolidation. This demonstrates that BS-CAT provides more comprehensive Dishonour capture.

4.2 Dishonours Consistency

Dishonours can be further broken down into lender-related and non-lender-related categories. A lender-related Dishonour is fundamentally a failed repayment and must be accurately assigned to the corresponding loan stream.

Among comparable samples where both parties identified the transaction as a Dishonours and assigned a specific counterparty, **the counterparty agreement rate is 95.2%**. The remaining **4.8%** discrepancy is concentrated in non-lender counterparties—for example, Dishonours transactions involving YOU! Insurance, Suncorp Insurance, Bupa, Telstra, etc. BS-CAT also flagged these as Dishonours but did not assign a specific lender counterparty, because the debit initiators in these cases are themselves non-lender entities and are therefore excluded from the loan-stream assignment consistency comparison.

In addition, for cases where Illion’s counterparty labelling is vague or imprecise, BS-CAT further demonstrates a stronger capability to correctly attribute transactions to the underlying loan stream.

Text Sample	Illion Category	BS-CAT Category	Illion third_party	BS-CAT counterparty
Loan Pymt Dishonour	Dishonours	Dishonours	Generic Loan	CBA Loans
Return 18/02/26 Direct Debit 600996 QuickCash DT.53v3ti...	Dishonours	Dishonours	Moneyspot	Quickcash

Across all loans that contain Dishonours, the counterparties for the three stages—drawdown, regular repayment, and Dishonour—are completely consistent, with 100% loan provider matching. This confirms that the full lifecycle of a given loan is accurately and consistently attributed to the same institution, with no breaks in assignment. In practical terms, a customer’s liability lifecycle (drawdown → regular repayment → repayment failure) is continuous and fully traceable under the BS-CAT model, leaving no coverage gaps in the Liability module.